

Full Stack Hiring Task

About BlendNet.ai

[BlendNet](#), conceptualized within Microsoft Research India as a pioneering [project](#) to bring digital services to the last mile, was spun off in 2023 to scale the solution in India. At its core, BlendNet focuses on harnessing the power of generative AI tools to build cutting edge products that keep people at the forefront of technological innovation. One of its key initiatives includes developing AI-driven platform aimed at enhancing learning outcomes, employability, and skilling in India. This aligns with our commitment to making quality education more affordable and accessible.

About Role

- **Role:** Full Stack SDE (Intern)
- **Duration:** 2-3 months
- **Stipend:** 25K per month plus other benefits
- **Full Time Conversion:** Full time offers will be extended to strong performers.

Responsibilities

As an intern at our company, you will be tasked with a variety of responsibilities designed to harness your skills and help you grow professionally within our team-oriented environment. Here's what you'll be focusing on:

1. **Development and Maintenance:** You will be responsible for both front-end and back-end development. This includes developing new user-facing features with React and TypeScript, as well as enhancing the backend using Python Django.
2. **Cross-Platform Design:** Engage in the design and development of responsive web and mobile applications that are user-friendly. Adapt our applications to various screen sizes using Material-UI, ensuring a seamless experience across all platforms, with the backend support powered by Django and hosted on Azure.
3. **Collaboration:** Collaborate closely with UX/UI designers to translate design concepts into functional web applications. Regularly participate in code reviews and team meetings to discuss and implement best practices and solutions across the full stack, working with both frontend and backend technologies.
4. **Testing and Optimization:** Implement rigorous testing procedures to ensure the application is robust and bug-free. Optimize the application for maximum speed and scalability, including the technical feasibility of UI/UX designs and backend performance.
5. **Learning and Growth:** Stay updated with emerging technologies and advancements in both frontend (React, TypeScript) and backend (Python Django) development. Continuously improve your skills and knowledge across these technologies, leveraging Azure for deployment and management.

6. **Documentation:** Maintain clear and comprehensive documentation regarding application specifics, project progress, and other significant information to support team collaboration and continuity. This includes documenting both frontend features and backend APIs, as well as deployment strategies on Azure.

Hiring Task Details

Objective: Design a stock monitoring platform where a user can login and monitor stock information.

Technologies:

- **Python (Django)** for backend will be preferred but feel free to use other frameworks or languages for your backend.
- **React:** Base framework for building the user interface.
- **TypeScript:** Employ strict typing practices wherever possible, avoiding the use of the **any** type.
- **Material UI:** Utilize for component styling and functionality.

Requirements:

- The platform should allow users to create and manage their own watchlists of stock symbols (e.g., MSFT, GOOG).
- The platform should display a dashboard with the latest stock values of the symbols on the user's watchlist.
- The platform should be able to handle multiple users concurrently, each having different watchlists.
- The platform should use a DB of your choice (e.g., MySQL/PostgreSQL/MongoDB) to store the user and watchlist data.
- The platform should use a secure and simple authentication mechanism for the users.
- The platform should use <https://www.alphavantage.co> as an API to pull stock information. The dashboard should show the latest stock price as returned by the `TIME_SERIES_INTRADAY` endpoint.

What we expect:

- We prefer that you code this in a Python or Java. The question is intentionally simple enough that you don't spend a lot of time in case you need to learn a new framework.
- We expect well written and structured APIs that follow good coding practices and standards.
- We expect a basic functional and intuitive UI that provides a good user experience.
- We expect working back-end code with sufficient documentation that explains the design choices and implementation details.
- We expect a link to a GitHub repo with the code and source files.

- The repo should have a detailed Readme.md file that captures the setup instructions for local deployment OR you can deploy the application on a cloud of your choice and share the URL. (You can consider platforms with free trial subscriptions like Azure, gCloud, Heroku etc)

Submission Note

It is acceptable if not all features are completely implemented within the given timeframe. If you are close to completing the task, we highly recommend submitting your work for review.

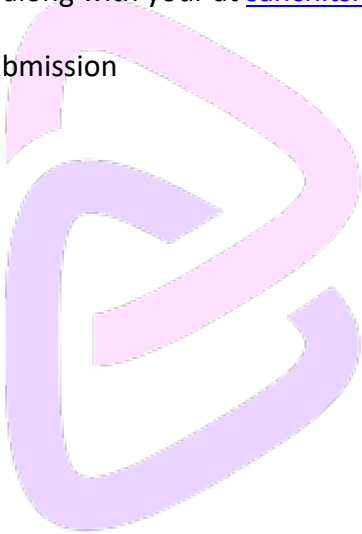
You can mail the following details along with your at sanchitsharma@blendnet.ai

Subject: [Full Stack] Hiring Task Submission

Your CV

Assignment GitHub Link

Assignment Deployment Link



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